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Palaeomagnetism of the Fishnish dykes, Mull, Scotland

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Summary. In Fishnish, eight dykes of E–W trend are all normally magnetized and all cut dykes of other trends. Normally magnetized dykes form a small minority of the dykes in the British Tertiary igneous province and this is the first reported incidence of a clearly defined group of a single polarity. The balance of the evidence – which is conflicting – favours the E–W dykes being contemporary with the later parts of the centre.

1 Sampling

The Fishnish peninsula on the northeast of Mull (Fig. 1) is intruded by a number of dykes, among which there are several cutting relationships. It is found that dykes of approximately E–W trend are later than dykes of any other trend (roughly N–S) that they intersect (Bailey 1924; Sloan 1971). All the dykes cut lavas of the plateau basalt sequence. Fifteen units (14 dykes and one inclined sheet) were sampled along the shore between NM 647423 and 651427 (OS 1 in. sheet 45) where Sloan (1971) has mapped 18 dykes. The approximate locations and trends of the dykes are shown in Fig. 1 and tentative correlations with Sloan's numbering is given in Table 1. It is not impossible that 105E or 106E is the same dyke as 109E or 110E.

To reduce the possibility of thermal re-magnetization, the older dyke at any intersection was sampled at least two dyke widths of the later dyke away from the contact. Af demagnetization was carried out for four cores from each dyke in steps of 5 millitesla (mT) up to at least 25 mT; the steps were continued, in some cases up to 140 mT, until either it was clear that a stable end point had been reached or was not attainable. At least two cores were thermally cleaned up to the Curie temperature, using 50°C heating steps, to compare with the af results.

Measurements of Curie point and magnetic susceptibility were also made for each core.

2 Results

Internal agreement between cores following af demagnetization was good except for 104E. The directions of magnetization after thermal and magnetic cleaning agreed well except for

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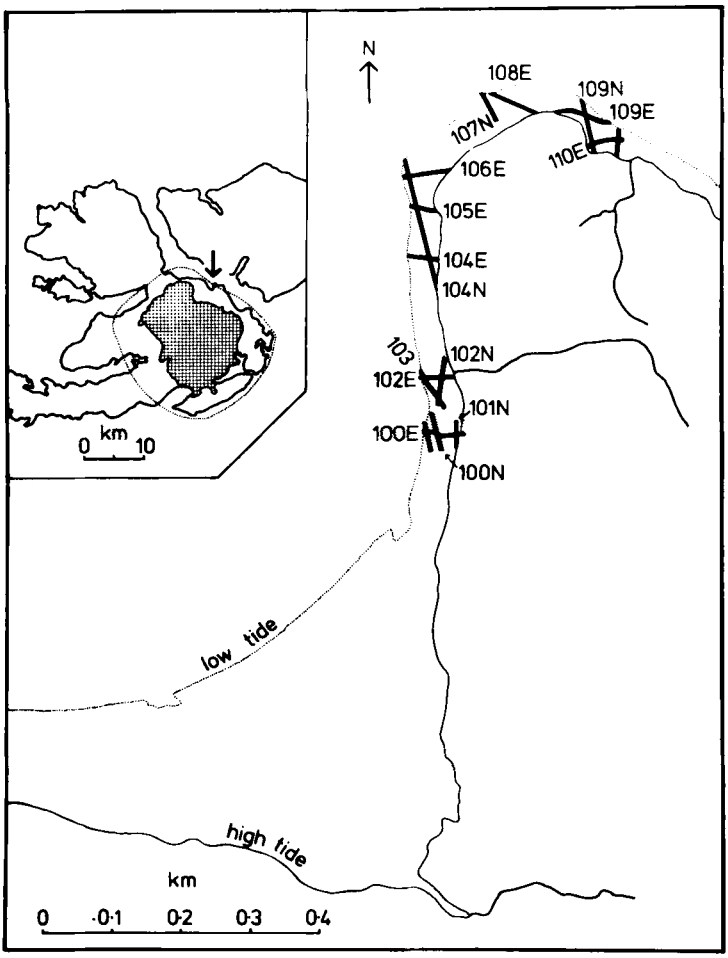


Figure 1. Map of the Fishnish peninsula, Isle of Mull showing the approximate location and trend of the dykes. Inset shows the location of the peninsula, the central intrusive complex (shaded) and the limit of the zone of pneumatolysis (dotted line).

dykes 101N, 104N and 109N. In these cases thermal cleaning to the Curie point (determined by rapid heating) did not remove all the magnetization, showing that chemical changes had occurred and therefore the magnetically-cleaned direction has been used.

Mean directions calculated for the af data using a 'minimum scatter' method (Dagley & Ade-Hall 1970) are given in Table 1 and Fig. 2. All eight E–W dykes are normally magnetized, while of the six roughly N–S dykes only one (107N) was normal the others being either reversed or intermediate with negative inclination (shallow reversed). The inclined sheet 103 also has an intermediate direction of magnetization. On the basis of mean directions 106E could be the same dyke as 109E.

The normally magnetized dykes have mean directions which cluster closely together and lie fairly close to the axial dipole field for Mull ($D = 0°, I = +71.7°$). The reversed directions are more widely dispersed and are not close to the dipole field direction. Almost all the cores from the reversely magnetized bodies had 'normal' remanance before cleaning, indicating a strong secondary component of magnetization which needed fields varying between 5 and 60 mT for its removal.

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Table 1. Palaeomagnetic results and cutting relationships for the Fishnish dykes (latitude 56.5° longitude 354.2°).

Dyke no.	Thickness	Strike °E of N	Direction of magnetization N	D(°ETN)	I° down		Polarity		Sloan's data Dyke no.	Thickness	Strike
100E	0.65	97–277	4	21.7	64.9	2.6	N	cuts 100N	F14	0.60	94–274 cuts F15
100N	2.4	166–346	4	202.9	−57.4	3.2	R	cut by 100E	F15	2.65	6–186 cut by F14
101N	1.0	0–180	4	175.6	−30.4	6.7	I (Sh. R)	cut by 100E	F5	0.90	20–200 cut by F14
102E	1.9	90–270	4	23.2	76.2	3.7	N	cuts 102N, 103	? F13	1.50	80–260 cuts F5, F9
102N	1.6	9–188	4	201.5	−53.4	7.1	R	cuts by 102E, 103			
103	0.93	145–325	3	183.4	−28.1	6.1	I (Sh. R)	cuts 102N, cut by 102E			
104E	1.7	100–280	4	226.1	79.1	10.0	probably N	cuts 104N	F10	1.85	104–284 cuts F5 and F9
104N	3.5	165–345	4	187.8	−33.4	6.6	R	cut by 104E, 105E, 106E	F9	4.2	178–358 cut by F14
105E	1.3	92–272	4	22.6	71.1	4.5	N	cuts 104N	F8	1.15	86–266 cuts F5 and F9
106E	1.0	80–260	4	7.3	73.8	2.6	N	cuts 104N	? F6	1.55	96–276 cuts F5 and F9
									or F8	1.15	86–266
107N	1.4	153–333	4	357.3	62.8	5.5	N		? FN29	1.20	154–334
108E	1.6	115–295	4	20.5	66.9	9.2	N		? FN25	1.65	168–348
109E	1.55	100–280	4	11.2	72.6	1.7	N	cuts 109N	? FN19	0.75	82–262 cuts FN21, F22, FN23
109N	2.4	165–345	3	212.7	−44.1	8.2	R	cut by 109E, 110E	? FN22	1.95	166–346 cut by FN19, FN20
110E	1.0	80–260	3	16.0	78.6	2.8	N	cuts 109N	FN20	1.20	78–258 cuts FN21, FN22, FN23

N = number of independently oriented cores used to establish mean direction.

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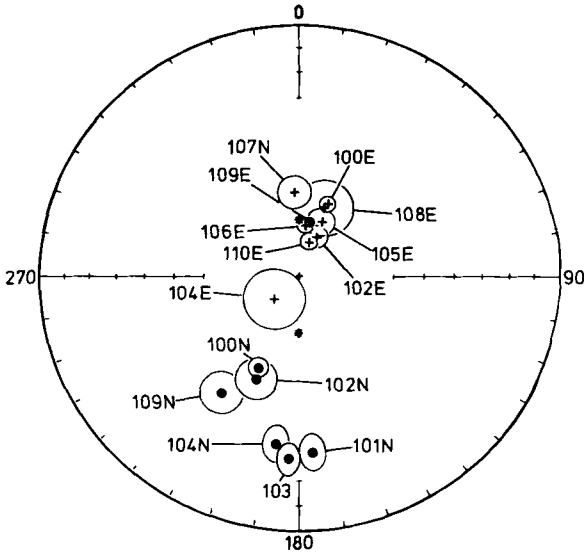


Figure 2. Equal area stereographic projection showing the unit mean directions of magnetization and circles of confidence (α_{95}) for the collection. * Indicates the direction of a centred axial dipole field.

Curie temperatures varied from 375 to 570°C with no significant systematic difference between N–S trending and E–W trending dykes. Only two dykes (101N and 102N) showed any indication of double Curie points (350 and 430°C respectively).

3 Discussion and conclusion

The palaeomagnetic directions of many dykes belonging to the N–W regional dyke swarms of the British Tertiary igneous province have been measured and in all the majority are reversely magnetized: 81 per cent in N. Syke (Wilson *et al.* 1974); 73 per cent in N. Mull (Ade-Hall *et al.* 1972); 92 per cent in Lundy (Mussett, Dagley & Eckford 1976), 78 per cent in Arran (unpublished work of this laboratory). In none of these cases has the magnetic polarity been found to correlate with any other property and the present study is the first for the province in which a clearly defined group of dykes has been shown to have a single polarity.

Two questions have to be asked: (a) Are the directions of magnetization primary? (2) What is their significance?

It is possible that the dykes have been remagnetized by the heat of later igneous bodies, particularly that of the central complex. Though Bailey (1924, p. 106) put the dykes just outside the zone of pneumatolysis, a zone of alteration produced by circulating hot fluids, Walker (1970) has shown that they lie in the prehnite zone of zeolitization that surrounds the complex like an annulus. Ade-Hall, Palmer & Hubbard (1971) give the lower stability limit of prehnite as at least 300°C so the dykes may have been heated somewhat above this temperature, but hardly above their Curie points, so they are unlikely to have been completely thermally remagnetized.

Regardless of this point it is difficult to see how remagnetization – either thermal or chemical – can have produced rocks of both polarities without there being periods of reversed and normal polarities. Furthermore, since the N–S dykes are earlier than the E–W ones they can hardly have experienced less remagnetization due to the heating and therefore their reverse magnetization was acquired earlier than the normal magnetization of the

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E–W ones. Only self-reversal or subtle differences of magnetic properties between the two sets of dykes can obviate this conclusion. Thus it is reasonable to conclude that for both sets, in spite of alteration, their primary remanance has not been lost and that the N–S dykes are typical members of the regional dyke swarms which contain both normally and reversely magnetized dykes (Wilson *et al.* 1974; Ade-Hall *et al.* 1972).

The N–S dykes are probably members of the Skye swarm which swing around through Morvern rather than of the main Mull swarm (Sloan 1971; Skelhorn 1969). However, both swarms have similar ages (Sloan, Speight & Skelhorn 1969), and their intrusion continued throughout most of the life of the centres. Although the Mull swarm considered as a whole is largely contemporaneous with the plateau lavas (Sloan 1971; Skelhorn 1969), which are all reversely magnetized (Hall, Wilson & Dagley 1977), this particular group of dykes must be younger than the lavas they intrude (60 ± 2 Ma) (Mussett *et al.* 1973).

The E–W dykes are clearly younger than both the lavas and the N–S dykes. Sloan (1971) has suggested a local origin associated with the Killbeg ring dyke which is one of the later major intrusions of the Mull centre (Skelhorn 1969). If this is so none of the N–S dykes which they cut are younger than the Mull centres and the normal E–W dykes must have roughly the same age as the centre. A Rb–Sr isochron age of 60.4 ± 1.9 Ma has been obtained for a granophyre of the last centre (Beckinsale 1974) and the youngest member of the central complex, the Loch Bá ring-dyke, is cut by reversely magnetized dykes, thought to belong to the NW–SE regional swarm (Ade-Hall 1974). The polarity sequence at 60 ± 2 Ma is then R → N → R with the plateau lavas during the first reversed period, the later part of the centre and the E–W dykes during the normal period and NW–SE regional dykes throughout.

This interpretation is not supported by a K–Ar date of 52 Ma (no details given) (Macintyre 1973) for the E–W Fishnish dykes which would imply that they were intruded during a normal period long after the emplacement of the Centre. An alternative explanation for the E–W dykes has been advanced by Holgate (1969) who associated them with a reversed stress regime in the area resulting from a dextral movement on the Great Glen Fault. He inferred this movement from a supposed offset of the Mull and Skye dyke swarms across the fault. This interpretation, unlike Sloan's, attributes the presence and E–W orientation of dykes to large-scale regional stress changes and presumably it would take a considerable time for such changes in the stress regime to come about. Thus the dykes would be significantly younger than the rest of the Mull Tertiary activity in line with the available K–Ar dates.

However, a reversal of the stress regime seems unlikely on a regional scale for dykes of NW–SE trend persisted until at least 52 Ma; dykes of this trend cut the Lundy granite 52.7 ± 1 Ma, (Dodson & Long 1962; Miller & Fitch 1962) and the Red Hills of Skye (~ 52 Ma old, Long 1964; Moorbath & Bell 1965). Moreover, careful mapping shows that the dyke swarms are not offset (Speight 1972) and this removes the basis of Holgate's hypothesis.

The unusual trend of the E–W dykes makes a local origin likely and so Sloan's interpretation is favoured although the reported dates are not consistent with an age contemporary with the centres.

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